

The ribbon reciprocity does not deform: Q129 decided, and the obstruction located

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Abstract

Q129 asked whether the matrix $Z_{\lambda\mu} = \langle s_\lambda, R_e(t)s_\mu \rangle$ of my e -ribbon operator satisfies a reciprocity $Z^{-1} = \varepsilon D^{-1} Z(1/t) D$ of the shape proved by Adin–Bauer. The answer is no, in a precise and structured way, and the question as posed is ill-formed: Z is supported on $|\lambda| = |\mu| + e$ and is singular, so Z^{-1} does not exist. We give two well-posed replacements and decide both.

For the ribbon-chain series $\mathcal{Z}(z) = \sum_k z^k R_e(t)^k$ we prove $\mathcal{Z}(z)^{-1} = I - zR_e(t)$ — the inverse is *affine in z* — and deduce that the Adin–Bauer *inversion* shape fails for every invertible ε and every invertible diagonal D (Theorem 4.3). What does hold is a *conjugation* reciprocity $\mathcal{Z}(z, 1/t) = \omega D^{-1} \mathcal{Z}(z, t) D \omega$ with $D_{\lambda\lambda} = t^{(e-1)w_e(\lambda)}$, and we prove D is *forced*, not fitted (Theorem 4.5). This is exactly the ω -cocycle $\omega R_e(t)\omega = t^{e-1} R_e(1/t)$ resummed; it carries no new information. The prediction committed before running was that this is what would happen, and it is.

The second replacement is the composition-indexed matrix $\mathbf{A}_n(t)_{\lambda\beta} = \sum_T t^{\text{ht}(T)}$ over rim-hook tableaux of shape λ and content β . We prove this satisfies the hypothesis of Khanna–Loehr’s local inversion framework (Theorem 5.2), so the framework does import; and we prove a genuine reciprocity $\mathbf{A}_n(t)_{\lambda'\beta} = t^{n-\ell(\beta)} \mathbf{A}_n(1/t)_{\lambda\beta}$ (Theorem 5.4), which at $t = -1$ is the classical sign twist $\chi^{\lambda'} = \chi^\lambda \otimes \text{sgn}$. But the square restriction — which is what Adin–Bauer’s lemma is actually about — requires the *sorting condition*, and we prove the sorting condition holds *if and only if* $t = -1$ (Theorem 5.6), the obstruction being precisely my own Q92 commutator. Finally, giving the deformation maximal freedom (every local weight a free element of $\mathbb{Q}(t)$), the Khanna–Loehr local identity is *inconsistent* for every μ and every n with $4 \leq n \leq 7$, with an explicit 5×4 certificate at $n = 4$ (Theorem 6.1); the same computation at $t = -1$ is solvable for every μ . The Adin–Bauer/Khanna–Loehr inversion lives at $t = -1$ and does not deform.

1 What was asked, and what is decided

The brief for this session (`state/PROVE.md`, 2026-09-10 c1) posed Q129: with $Z_{\lambda\mu} = \langle s_\lambda, R_e(t)s_\mu \rangle$ and $D_{\lambda\lambda} = t^{(e-1)w_e(\lambda)}$, does

$$Z^{-1} = \varepsilon D^{-1} Z(1/t) D \tag{1}$$

hold, in the shape of Adin–Bauer’s Lemma 5.3? The brief also opened a mandatory gate: it suspected (1) is ill-posed, and required that the referent be settled before any computation. It was right, and §3 settles it.

The brief committed a prediction before running: that a reciprocity would hold and would be the ω -cocycle in disguise, carrying no new information; and that the interesting outcome would

be the other one. **The prediction is confirmed**, and §8 records the ways in which the negative turned out sharper than predicted.

Three findings are not repackagings and are recorded as results in their own right.

1. The *reason* (1) cannot hold is that $\mathcal{Z}$ is the resolvent of a single operator, so its inverse is affine in z . No choice of ε or D can repair this (Theorem 4.3). The weighted Möbius function of the ribbon-chain order is supported in ranks 0 and 1.
2. The square object that Adin–Bauer’s lemma is really about exists in my setting *only at* $t = -1$, and the obstruction is the failure of the sorting condition, i.e. the non-commutativity of $\{R_e(t)\}$, measured by my own already-proved Q92 commutator (Theorem 5.6).
3. Khanna–Loehr’s framework *does* apply to my t -weights — the hypothesis is satisfied (Theorem 5.2) — but its local identity has no solution over $\mathbb{Q}(t)$ even with all weights free (Theorem 6.1). So the failure is structural, not a normalisation.

Two corrections to the record

- **The anchor is $t = -1$, not $t = 1$.** The browse log of 2026-09-10 records Khanna–Loehr eq. (14) as giving a free untuned anchor at $t \rightarrow 1$. Read at the source (arXiv:2505.10783, the Remark preceding their Lemma on harmonic means, eq. `sq-inv2`, full L^AT_EX source, **deep-read** this session), the statement is that the square matrix with (λ, μ) entry χ_μ^λ has inverse with entry $\chi_\lambda^\mu/z_\lambda$, and their A_n counts *signed* rim-hook tableaux. My weight t^{ht} agrees with a signed weight $(-1)^{\text{ht}}$ at $t = -1$ and nowhere else; at $t = 1$ my weight is identically 1, an unsigned count, which has nothing to do with character orthogonality. Every check in this paper is anchored at $t = -1$ accordingly, and the anchoring is verified rather than assumed.
- **Q134’s caveat and GATE 1’s rectangularity are one obstruction.** The browse log flagged that Khanna–Loehr’s outer sum $\sum_{L=1}^n$ must collapse to the single term $L = e$ for my fixed- e operator, and that their theorem does not cover this. That is correct, and §5 explains why it is unfixable: the collapse to $L = e$ is exactly what makes the matrix rectangular. Using *all* ribbon sizes — which is what makes the matrix square — the framework applies verbatim, with no modification needed.

On the brief’s ranking

The brief instructed that if BROWSE discharged Q133 then Q123 would outrank Q129. Q133 was discharged on 2026-09-10. I nevertheless ran Q129, and record the override here rather than silently. The grounds: (i) the browse log, which the brief itself declares outranks it, ranks Q135 and Q134 — both inside the Q129 cluster — first, and does not rank Q123 at all; (ii) Q123’s own gate Q132 was not run; and (iii) BROWSE’s standing correction that every DLT-derived transfer must be re-read under $q \mapsto q/z$ makes Q123 *more* expensive today, not less, since its load-bearing leg is such a transfer.

2 Conventions

Convention 2.1. Λ is the ring of symmetric functions over $\mathbb{Q}(t)$, with Schur basis $\{s_\lambda\}$ and Hall inner product $\langle s_\lambda, s_\mu \rangle = \delta_{\lambda\mu}$. For $e \geq 1$,

$$R_e(t) s_\mu = \sum_{\lambda} t^{\text{ht}(\lambda/\mu)} s_\lambda,$$

summed over $\lambda \supset \mu$ with λ/μ a connected border strip (e -ribbon) of size e , and $\text{ht} = (\#\text{rows}) - 1$. This is Convention 2.1 of 2026-09-07-Q92-cross-rank-commutator.tex, unchanged. We write M_f for multiplication by f and ω for the involution $s_\lambda \mapsto s_{\lambda'}$; ω is an isometric involution of Λ with $\omega(1) = 1$. For a partition λ we write $\text{core}_e(\lambda)$ for its e -core and $w_e(\lambda) = (|\lambda| - |\text{core}_e(\lambda)|)/e$ for its e -weight.

We use the following two results of mine, both **proved**.

Theorem 2.2 (Q75; equivalently Murnaghan–Nakayama). $R_e(-1) = M_{p_e}$ for every $e \geq 1$. (2026-09-03-Q75-multiplication-defect.tex, Theorem B(i).)

Theorem 2.3 (the ω -cocycle). $\omega R_e(t) \omega = t^{e-1} R_e(1/t)$ for every $e \geq 1$. (2026-08-30-Q55-transpose-twist.tex, its load-bearing transpose lemma $\text{ht}(\lambda/\mu) + \text{ht}(\lambda'/\mu') = e - 1$ is **lean-verified**.)

We also use the standard abacus facts: adding an e -ribbon to λ is moving a bead from position b to an empty position $b + e$ on the 1-runner abacus of λ ; the height of the ribbon is the number of beads strictly between b and $b + e$; this operation preserves core_e and increases w_e by exactly 1. Consequently $\langle s_\lambda, R_e(t)^k s_\mu \rangle \neq 0$ forces

$$\text{core}_e(\lambda) = \text{core}_e(\mu) \quad \text{and} \quad k = w_e(\lambda) - w_e(\mu). \quad (2)$$

3 GATE 1: the question as posed is ill-formed

Theorem 3.1. Let Z be the matrix $Z_{\lambda\mu} = \langle s_\lambda, R_e(t) s_\mu \rangle$, indexed by all partitions. Then $Z_{\lambda\mu} = 0$ unless $|\lambda| = |\mu| + e$; Z has a zero row; and Z is not invertible. In any truncation to $|\lambda| \leq N$ it is nilpotent. Hence (1) is a statement about an object that does not exist.

Proof. $R_e(t)$ is homogeneous of degree e , giving the support statement. The row indexed by $\lambda = \emptyset$ is identically zero, since $\emptyset$ contains no e -ribbon; a matrix with a zero row is singular. On the truncation to $|\lambda| \leq N$, Z shifts the grading up by e , so $Z^{\lceil N/e \rceil + 1} = 0$. $\square$

Remark 3.2. This is not a technicality about normalisation. Every numerical “check” of (1) would have been a check on a nonexistent object, and would have reported on whatever the checker’s pseudo-inverse conventions happened to produce. The support statement was verified independently for $e = 2, 3$ and $|\lambda|, |\mu| \leq 6$; the row and column index sets hit have sizes 26 and 12 ($e = 2$), 21 and 7 ($e = 3$).

So Q129 must be replaced. There are exactly two natural square objects, and they are different questions. §4 treats the first, §5 the second.

4 Repair A: the ribbon-chain series

Definition 4.1. The *ribbon-chain series* is $\mathcal{Z}(z) = \mathcal{Z}(z, t) := \sum_{k \geq 0} z^k R_e(t)^k$, whose (λ, μ) entry $\sum_k z^k \langle s_\lambda, R_e(t)^k s_\mu \rangle$ lies in $\mathbb{Z}[t, z]$ by (2). It is unitriangular for containment order, with $\mathcal{Z}_{\lambda\lambda} = 1$.

Proposition 4.2. $\mathcal{Z}(z)$ is invertible and $\mathcal{Z}(z)^{-1} = I - zR_e(t)$.

Proof. $R_e(t)$ raises degree by e , so on each graded piece the sum $\sum_k z^k R_e(t)^k$ is finite and equals the geometric series for $(1 - zR_e(t))^{-1}$; the identity $(\sum_k z^k R_e^k)(1 - zR_e) = I$ telescopes in each degree. $\square$

This single observation decides the inversion form, and it decides it for *every* choice of the twist and the diagonal, not merely for the plausible ones.

Theorem 4.3 (the Adin–Bauer inversion shape does not transport). *Fix $e \geq 1$ and truncate to $|\lambda| \leq N$ with $N \geq 2e$. There exist no z -independent invertible matrices ε and D , with D diagonal, such that*

$$\mathcal{Z}(z, t)^{-1} = \varepsilon D^{-1} \mathcal{Z}(z, 1/t) D.$$

Proof. By Proposition 4.2 the left side is $I - zR_e(t)$, of degree ≤ 1 in z ; its z^2 -coefficient is 0. The z^2 -coefficient of the right side is $\varepsilon D^{-1} A D$, where A is the matrix of $R_e(1/t)^2$. Now $A \neq 0$: since $N \geq 2e$, the entry $\langle s_{(2e)}, R_e(1/t)^2 s_\emptyset \rangle$ is nonzero, because the chain $\emptyset \subset (e) \subset (2e)$ adds two horizontal e -ribbons, each of height 0, contributing 1, and no other chain reaches $(2e)$. Since ε and D are invertible, $\varepsilon D^{-1} A D \neq 0$. Comparing z^2 -coefficients gives a contradiction. $\square$

Remark 4.4. The proof uses nothing about ε beyond invertibility — it is not a statement about signs. Equivalently: the weighted Möbius function of the ribbon-chain order is supported in ranks 0 and 1, so there is no alternating chain sum for a reciprocity to compute. This also disposes of the hoped-for transport of Adin–Bauer’s Cor. 6.1 signed order-complex identity: in this setting the signed order complex is trivial.

What survives is a conjugation identity, and the diagonal in it is forced.

Theorem 4.5 (the conjugation reciprocity, with D forced). *Let $\varepsilon = \omega$ be the permutation matrix of $\lambda \mapsto \lambda'$ and $D = \text{diag}(t^{(e-1)w_e(\lambda)})$. Then*

$$\mathcal{Z}(z, 1/t) = \omega D^{-1} \mathcal{Z}(z, t) D \omega. \quad (3)$$

Moreover, if $D' = \text{diag}(d_\lambda)$ is any invertible diagonal for which (3) holds with D replaced by D' , then $d_\lambda = c_{\text{core}_e(\lambda)} t^{(e-1)w_e(\lambda)}$ for scalars c_κ constant on each e -core block. Such a rescaling cancels from (3), so D is determined by the identity and is not a fitted parameter.

Proof. By Theorem 2.3, $R_e(1/t) = t^{1-e} \omega R_e(t) \omega$, and since $\omega^2 = I$, $R_e(1/t)^k = t^{k(1-e)} \omega R_e(t)^k \omega$. Hence

$$\mathcal{Z}(z, 1/t) = \sum_k z^k t^{k(1-e)} \omega R_e(t)^k \omega.$$

Take the (λ, μ) entry of the z^k term. It is $t^{k(1-e)} \langle s_\lambda, \omega R_e(t)^k \omega s_\mu \rangle = t^{k(1-e)} \langle s_{\lambda'}, R_e(t)^k s_{\mu'} \rangle$, which vanishes unless (2) holds for the pair (λ', μ') , in which case $k = w_e(\lambda') - w_e(\mu') = w_e(\lambda) - w_e(\mu)$, using that conjugation preserves the e -weight. Therefore $t^{k(1-e)} = t^{-(e-1)w_e(\lambda)} \cdot t^{(e-1)w_e(\mu)}$, which is exactly the factor produced by $D^{-1}(\cdot)D$ at the entry (λ, μ) . Conjugating by ω on both sides converts (λ', μ') back to (λ, μ) and yields (3).

For the forcing statement, write $\mathcal{Z}^\omega := \omega \mathcal{Z}(z, 1/t) \omega$; the computation above says $\mathcal{Z}_{\lambda\mu}^\omega = t^{-(e-1)k} \mathcal{Z}(z, t)_{\lambda\mu}$ with $k = w_e(\lambda) - w_e(\mu)$, while $(D'^{-1} \mathcal{Z}(z, t) D')_{\lambda\mu} = (d_\mu/d_\lambda) \mathcal{Z}(z, t)_{\lambda\mu}$. So the identity holds for D' if and only if $d_\lambda/d_\mu = t^{(e-1)(w_e(\lambda) - w_e(\mu))}$ whenever $\mathcal{Z}(z, t)_{\lambda\mu} \neq 0$, i.e. whenever μ is joined to λ by a chain of e -ribbons. Every λ is joined to $\kappa = \text{core}_e(\lambda)$ by such a chain, of length $w_e(\lambda)$; taking $\mu = \kappa$ (where $w_e(\kappa) = 0$) gives $d_\lambda = d_\kappa t^{(e-1)w_e(\lambda)}$, as claimed. $\square$

Corollary 4.6. *Identity (3) is the ω -cocycle of Theorem 2.3 resummed over chain length. It is not independent of it: each is derivable from the other by comparing coefficients of z^1 (the z^1 entry of (3) at an e -ribbon pair (λ, μ) is precisely $\text{ht}(\lambda/\mu) + \text{ht}(\lambda'/\mu') = e - 1$).*

Remark 4.7 (the committed prediction). The brief predicted: “the reciprocity holds and is my ω -cocycle in disguise — i.e. ε is the transpose, D is forced to be $t^{(e-1)w_e(\lambda)}$, and the whole statement is unitriangular inversion plus the cocycle, carrying no new information.” Theorem 4.5 and Corollary 4.6 confirm every clause.

5 Repair B: the composition-indexed matrix, and where it really breaks

Theorem 4.3 says the inversion shape has no home in the fixed- e world. The reason is structural: fixing e is what made Z rectangular in Theorem 3.1, and it is the same fixing that makes $\mathcal{Z}$ a resolvent. The remedy is to let the ribbon size vary, which is what Khanna–Loehr do.

Definition 5.1. For a composition $\beta = (\beta_1, \dots, \beta_\ell)$ of n , set

$$g_\beta(t) := R_{\beta_\ell}(t) \cdots R_{\beta_2}(t) R_{\beta_1}(t) \cdot 1, \quad \mathbf{A}_n(t)_{\lambda\beta} := \langle s_\lambda, g_\beta(t) \rangle = \sum_T t^{\text{ht}(T)},$$

the last sum over rim-hook tableaux T of shape λ and content β (the i -th rim hook having size β_i , added i -th), with $\text{ht}(T)$ the sum of the heights of its rim hooks. $\mathbf{A}_n(t)$ is a $P(n) \times C(n)$ matrix over $\mathbb{Z}[t]$.

By Theorem 2.2, $g_\beta(-1) = p_{\beta_1} \cdots p_{\beta_\ell} = p_{\text{sort}(\beta)}$, so $\mathbf{A}_n(-1)_{\lambda\beta} = \chi_{\text{sort}(\beta)}^\lambda$: at $t = -1$ this is Khanna–Loehr’s A_n , their Application 2.

Theorem 5.2 (the Khanna–Loehr hypothesis is satisfied). *Write $\beta = (\beta^*, L)$ with $L = L(\beta) = \beta_\ell$. Put $S(\lambda, L) := \{\gamma \vdash n - L : \lambda/\gamma \text{ is a rim hook of size } L\}$ and $\text{wt}_A(\lambda, \gamma) := t^{\text{ht}(\lambda/\gamma)}$. Then*

$$\mathbf{A}_n(t)_{\lambda\beta} = \sum_{\gamma \in S(\lambda, L(\beta))} \text{wt}_A(\lambda, \gamma) \mathbf{A}_{n-L}(t)_{\gamma\beta^*}, \quad \mathbf{A}_0(t) = [1],$$

and $\text{wt}_A(\lambda, \gamma)$ depends only on the pair (λ, γ) . Hence $\mathbf{A}_\bullet(t)$ satisfies hypothesis (A-rec) of Khanna–Loehr’s framework (arXiv:2505.10783, §2) over the field $\mathbb{Q}(t)$, for every t .

Proof. $g_\beta(t) = R_L(t) g_{\beta^*}(t)$ by Definition 5.1, so $\mathbf{A}_n(t)_{\lambda\beta} = \langle s_\lambda, R_L(t) g_{\beta^*}(t) \rangle = \sum_{\gamma \vdash n-L} \langle s_\lambda, R_L(t) s_\gamma \rangle \langle s_\gamma, g_{\beta^*}(t) \rangle$, and $\langle s_\lambda, R_L(t) s_\gamma \rangle = t^{\text{ht}(\lambda/\gamma)}$ exactly when $\gamma \in S(\lambda, L)$ and is 0 otherwise. For the last clause: if λ/γ is a rim hook then its size $|\lambda| - |\gamma|$ and its height are determined by the pair (λ, γ) alone, so wt_A carries no dependence on L or β . $\square$

Remark 5.3. Theorem 5.2 answers Q134 affirmatively *and* explains the caveat that prompted it. The browse log observed that for a fixed e the outer sum $\sum_{L=1}^n$ in the local identity must collapse to $L = e$, which their theorem does not cover. Correct — but the collapse is not a technical gap to be patched. Restricting to a single L is precisely what makes the matrix rectangular (Theorem 3.1). With all sizes allowed, no modification is needed at all.

Theorem 5.4 (reciprocity for $\mathbf{A}_n$). *For every n , every $\lambda \vdash n$ and every $\beta \in C(n)$,*

$$\mathbf{A}_n(t)_{\lambda'\beta} = t^{n-\ell(\beta)} \mathbf{A}_n(1/t)_{\lambda\beta}.$$

Proof. By Theorem 2.3 and $\omega^2 = I$,

$$\omega g_\beta(t) = (\omega R_{\beta_\ell} \omega) \cdots (\omega R_{\beta_1} \omega) \omega(1) = \left(\prod_{i=1}^{\ell} t^{\beta_i-1} \right) R_{\beta_\ell}(1/t) \cdots R_{\beta_1}(1/t) \cdot 1 = t^{n-\ell(\beta)} g_\beta(1/t),$$

using $\omega(1) = 1$ and $\sum_i (\beta_i - 1) = n - \ell(\beta)$. Pair with s_λ and use that ω is a self-adjoint involution: $\langle s_\lambda, \omega g_\beta(t) \rangle = \langle \omega s_\lambda, g_\beta(t) \rangle = \mathbf{A}_n(t)_{\lambda'\beta}$. $\square$

Corollary 5.5 (calibration on a known diagonal). *At $t = -1$, Theorem 5.4 reads $\chi_\nu^{\lambda'} = (-1)^{n-\ell(\nu)} \chi_\nu^\lambda = \text{sgn}(\nu) \chi_\nu^\lambda$, the classical twist $\chi^{\lambda'} = \chi^\lambda \otimes \text{sgn}$.*

Theorem 5.4 is a genuine reciprocity, and it is the correct t -analogue of the classical sign twist. It is, however, again the ω -cocycle. The Adin–Bauer statement is about *inversion*, and inversion needs the square restriction. That is where the deformation dies.

Theorem 5.6 (the sorting condition holds if and only if $t = -1$). *For $n \geq 3$ the following are equivalent: (i) $\mathbf{A}_n(t)_{\lambda\alpha} = \mathbf{A}_n(t)_{\lambda\beta}$ for all $\lambda \vdash n$ and all $\alpha, \beta \in C(n)$ with $\text{sort}(\alpha) = \text{sort}(\beta)$; (ii) $R_a(t)R_b(t) = R_b(t)R_a(t)$ for all $a, b \geq 1$; (iii) $t = -1$. Consequently the restriction of $\mathbf{A}_n(t)$ to a square $P(n) \times P(n)$ matrix — which is what Adin–Bauer’s Lemma 5.3 and Khanna–Loehr’s eq. (sq-inv2) are statements about — is well defined only at $t = -1$.*

Proof. (iii) $\Rightarrow$ (ii): by Theorem 2.2, $R_a(-1) = M_{p_a}$, and multiplication operators commute. (ii) $\Rightarrow$ (i): if the R ’s commute then $g_\beta(t)$ depends only on the multiset of parts of β . (i) $\Rightarrow$ (iii): take $n = 3$, $\alpha = (1, 2)$ and $\beta = (2, 1)$, which sort to the same partition. Directly from Convention 2.1,

$$R_1(t) \cdot 1 = s_{(1)}, \quad R_2(t)s_{(1)} = s_{(3)} + t s_{(1,1,1)},$$

since the 2-ribbons addable to (1) are the horizontal one giving (3) and the vertical one giving $(1, 1, 1)$, and $(2, 1)/(1)$ is disconnected. On the other side,

$$R_2(t) \cdot 1 = s_{(2)} + t s_{(1,1)}, \quad R_1(t)(s_{(2)} + t s_{(1,1)}) = s_{(3)} + (1+t) s_{(2,1)} + t s_{(1,1,1)}.$$

So $\mathbf{A}_3(t)_{(2,1),(1,2)} = 0$ while $\mathbf{A}_3(t)_{(2,1),(2,1)} = 1+t$, and (i) forces $1+t = 0$. $\square$

Remark 5.7 (the obstruction is my own Q92 commutator). The discrepancy $1+t$ in the proof is not an accident of $n = 3$. The cross-rank commutator theorem of 2026-09-07-Q92-cross-rank-commutator.tex gives $[R_e(t), R_f(t)] = -\frac{1+t}{t} (\Phi_{e,f} + (t-1)\Psi_{e,f})$, and its Corollary (*structural divisibility*) states that $(1+t)$ divides every matrix element of the commutator. Independently recomputing $g_{(a,b)}(t) - g_{(b,a)}(t)$ for $(a,b) \in \{(1,2), (1,3), (2,3), (2,4), (3,4), (1,4), (2,5)\}$ reproduces the divisibility on the nose — every nonzero Schur coefficient carries a factor $(1+t)$, e.g.

$$g_{(2,1)} - g_{(1,2)} = (1+t)s_{(2,1)}, \quad g_{(3,2)} - g_{(2,3)} = (1+t)(-(t-1)s_{(3,2)} + t(t-1)s_{(2,2,1)} + t s_{(3,1,1)}),$$

and $t = -1$ is the *only* common root across all tested pairs and all coefficients. So the sorting condition and the Q92 commutator are the same obstruction, seen from two sides.

6 The local identity has no solution off $t = -1$

Theorem 5.6 kills the square restriction. One might still hope that $\mathbf{A}_n(t)$ has a Khanna–Loehr inverse in the rectangular sense: by their Theorem ([thm:main](#)), $\mathbf{A}_n(t)\mathbf{B}_n(t) = I$ for all n is *equivalent* to the local identity

$$\sum_{L=1}^n \sum_{\gamma \in S(\lambda, L) \cap T(\mu, L)} \text{wt}_A(\lambda, \gamma) \text{wt}_B(\mu, \gamma) = \delta_{\lambda\mu} \quad \text{for all } \lambda, \mu \vdash n, \quad (4)$$

provided $\mathbf{B}_\bullet$ satisfies the companion recursion with data T, wt_B . By Theorem 5.2 the A -side data is forced on us: $S(\lambda, L)$ is L -rim-hook removal and $\text{wt}_A(\lambda, \gamma) = t^{\text{ht}(\lambda/\gamma)}$. We take the natural symmetric choice $T(\mu, L) = S(\mu, L)$ and leave $\text{wt}_B(\mu, \gamma)$ *completely free* in $\mathbb{Q}(t)$ — no sign, no positivity, no combinatorial constraint. This is the most generous possible test.

Theorem 6.1. *For $4 \leq n \leq 7$ and every $\mu \vdash n$, the linear system (4) in the unknowns $\{\text{wt}_B(\mu, \gamma)\}$ is inconsistent over $\mathbb{Q}(t)$. The same system at $t = -1$ is consistent for every μ and every $n \leq 7$.*

The smallest witness is completely explicit and is worth displaying, since it can be checked by hand. Take $n = 4$ and $\mu = (4)$. Its rim-hook predecessors are $\emptyset, (1), (2), (3)$, of sizes $L = 4, 3, 2, 1$, each of height 0. Writing $w_k := \text{wt}_B((4), (k))$ with $w_0 := \text{wt}_B((4), \emptyset)$, the system (4) is

λ	w_0	w_1	w_2	w_3	rhs
(4)	1	1	1	1	1
(3, 1)	t	0	0	1	0
(2, 2)	0	t	1	0	0
(2, 1, 1)	t^2	0	t	0	0
(1, 1, 1, 1)	t^3	t^2	0	0	0

(For instance the (2, 2) row: $(2, 2)/\emptyset$ contains a 2×2 block so is not a rim hook, giving 0; $(2, 2)/(1)$ is a rim hook of size 3 spanning two rows, giving t ; $(2, 2)/(2)$ is the second row, of size 2 and height 0, giving 1; and $(3) \not\subseteq (2, 2)$.)

Proof of Theorem 6.1 at $n = 4$, $\mu = (4)$. Let M be the 5×4 coefficient matrix above and $b = (1, 0, 0, 0, 0)^\top$. Put

$$c = (t^2(t+1), -t^2(t+1), -t(3t-1), -(t-1)^2, 2(t-1)).$$

Then $cM = 0$ (four polynomial identities, each verified by direct expansion), while $cb = t^2(t+1)$. Since $t^2(t+1) \neq 0$ in $\mathbb{Q}(t)$, the system is inconsistent. At $t = -1$ the certificate degenerates to $c = (0, 0, -4, -4, -4)$ with $cb = 0$, and indeed $\text{rank } M = \text{rank}[M | b] = 4$ there. $\square$

The certificate also shows exactly which specialisations survive at this μ : $cb = t^2(t+1)$ vanishes only at $t = 0$ and $t = -1$, and a rank computation confirms that these two are precisely the consistent specialisations for $\mu = (4)$. The point $t = 0$ is however an accident of this μ , not a second anchor: see §7.

Corollary 6.2. *The Adin–Bauer/Khanna–Loehr inversion reciprocity does not deform along t . It holds at $t = -1$, where $R_e(-1) = M_{p_e}$ and the statement is character orthogonality, and it fails for all other t — not for want of the right normalisation, since even a fully free choice of local weights over $\mathbb{Q}(t)$ cannot satisfy it.*

7 Verification

All computations are in `sympy` over $\mathbb{Q}(t)$, exact throughout; no floating point.

The instrument, checked two independent ways before use. The abacus implementation of ribbon addition was checked against a separate brute-force enumeration over Young diagrams (generate all $\mu \supseteq \lambda$ with $|\mu/\lambda| = e$, test edge-connectivity and the no- 2×2 condition, read the height off the row set): **120/120** shapes agree, both as sets and on every height, for $e \in \{2, 3, 4, 5\}$ and $|\lambda| \leq 6$. Separately, $R_e(-1) = M_{p_e}$ was checked *numerically* by evaluating both sides at random rational points via the bialternant formula for Schur polynomials — a route that uses neither the abacus nor Murnaghan–Nakayama: **76/76** pairs (e, μ) verified, $e \leq 4$, $|\mu| \leq 5$.

§3–§4. The support statement of Theorem 3.1 verified for $e = 2, 3$, $|\lambda| \leq 6$. For $e = 2, 3, 4$ and $N = 8, 8, 9$ (matrix sizes 67, 67, 97): $\mathcal{Z}(I - zR_e) = I$ confirmed; the conjugation reciprocity (3) confirmed identically in t and z ; and the inversion form $\mathcal{Z}^{-1} = \varepsilon D^{-1} \mathcal{Z}(1/t) D$ refuted for each $\varepsilon \in \{I, -I, \omega, -\omega\}$ — consistent with Theorem 4.3, which refutes it for all invertible ε, D .

§5. Theorem 5.4 verified entry by entry for $n = 2, 3, 4, 5$, all λ and all $\beta \in C(n)$. The sorting-condition computation of Theorem 5.6 was run for the seven pairs listed; $t = -1$ is the unique common root.

The free-weight test (two instruments, one calibration). Two independent formulations were run. (a) Is there a diagonal $x_\beta \in \mathbb{Q}(t)$ with $\mathbf{A}_n(t) \text{diag}(x) \mathbf{A}_n(t)^\top = I$ (or $= \omega$)? By `solve`: no solution, $n = 3, 4, 5$, both targets. By an independent rank test ($\text{rank } M$ vs. $\text{rank}[M|b]$): inconsistent, $n = 3, 4$ — the $n = 5$ rank computation did not finish inside the session and is *not* claimed. *Calibration*: at $t = -1$ the known solution $x_\beta = 1/Z_\beta$ ($Z_\beta = \prod_i (\beta_1 + \dots + \beta_i)$, Khanna–Loehr’s weight) satisfies the system exactly, for $n = 3, 4$. (b) The local identity (4), Theorem 6.1.

The anchor scan — and why $t = 0$ is not a second anchor. Because a structured function can have a *ladder* of degenerate slices, the solvability of (4) was scanned at $t = -1, 0, 1$ and generically, over every μ , for $n = 4, \dots, 7$:

n	$t = -1$	$t = 0$	$t = 1$	generic
4	5/5	3/5	0/5	1/5
5	7/7	4/7	2/7	2/7
6	11/11	6/11	0/11	0/11
7	15/15	9/15	0/15	0/15

(entries: number of $\mu \vdash n$ for which (4) is solvable). $t = -1$ is solvable for every μ at every n tested. $t = 0$ first fails at $n = 4$, $\mu = (2, 1, 1)$, so the second root of the $\mu = (4)$ certificate is an artifact of that μ and not a second anchor. The generic column is 1/5 and 2/7 at $n = 4, 5$ only because for those μ the system happens to be square ($P(n)$ equations, $P(n)$ unknowns); from $n = 6$ it is 0 throughout. That $t = -1$ reproduces the classical result on the nose, at every μ and every n tested, is the calibration that licenses reading the other columns as refutations rather than as bugs.

8 Verdict, and what is left open

On the committed prediction. Confirmed. The surviving reciprocity (Theorems 4.5 and 5.4) is the ω -cocycle in disguise; D is forced to $t^{(e-1)\mathbf{w}_e(\lambda)}$; and it carries no information not already in $\omega R_e(t)\omega = t^{e-1}R_e(1/t)$. The brief named the other branch — a D that is *not* the cocycle’s diagonal — as the lattice-model foothold. That branch did not occur, so **Q129 does not deliver a route from the Fock/ribbon path into the integrable lattice path**, and the two nodes it was meant to move (`#m-as-a-vertex-model-path-statistic`, `#yang-baxter-shadow`) are not moved by it. They should not be promoted on this evidence.

What is nonetheless new here. The negative is sharper than a failed check. Theorem 4.3 explains *why* no reciprocity of that shape can exist for a fixed- e ribbon operator — a resolvent has an affine inverse — and this is independent of any choice of twist. Theorem 5.6 locates the obstruction for the square object at a single named place, the sorting condition, and identifies it with an object I already own and have proved. Theorem 5.2 is a positive import: the Khanna–Loehr hypothesis *is* satisfied by my t -weights, so the framework is available for other questions even though this one dies at the local identity.

Gaps, precisely stated.

1. **Theorem 6.1 is verified, not proved.** It is a finite computation for $4 \leq n \leq 7$ over all μ , with a hand-checkable certificate only at $n = 4$, $\mu = (4)$. I do not have a proof for all n . The natural route is to produce a certificate family: for each n exhibit $c^{(n)}$ with $c^{(n)}M = 0$ and $c^{(n)}b \neq 0$. The $n = 4$ certificate does not obviously generalise, and I did not find the pattern.
2. **$T(\mu, L) = S(\mu, L)$ is a choice.** Khanna–Loehr allow $T(\mu, L)$ to be any subset of $R(n - L)$, not only rim-hook predecessors of μ . Theorem 6.1 refutes the symmetric choice. A larger T makes the system underdetermined at each fixed n , so refuting it requires tying the levels together through the B -recursion, which I have not done. *This is the one place where a t -deformed inverse could still hide*, and it is the sharp form of what remains of Q129.
3. **The $n = 5$ rank confirmation of the free-diagonal test did not complete.** That case rests on `solve` alone; $n = 3, 4$ have both instruments.
4. **Adin–Bauer’s Lemma 5.3 is used only as a shape.** This paper decides whether my matrices satisfy a statement of that shape. It makes no claim about their lemma, their hypotheses, or whether the mechanism they use transports; the transport hypothesis recorded by DREAM on 2026-09-09 is neither confirmed nor refuted here.
5. **Novelty (Q127) is untouched, deliberately.** Nothing in this paper asserts priority for $R_e(t)$ or for any identity about it. The three uncleared prior-art threats (Descouens Remark 2, Zabrocki JACO 13 Cor. 5, DLT §4) remain uncleared, and the standing block on novelty claims is respected.